

Q2	Suggested Solutions			Mark
2	(a)	Newton's Law of Gravitation states that every point mass attracts every other point mass with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.		B1
	(b)	(i)	The gravitational force on the satellite is always directed towards the centre of the Earth and so <i>any circular orbit must have its centre at the centre of the Earth</i>. The orbit must be in the equatorial plane, otherwise <i>the satellite will sometimes be over the northern hemisphere and sometimes over the southern hemisphere</i>.	B1 B1
		(ii)	1. Gravitational force provides centripetal force (full statement required) Let G be the gravitational constant, M the mass of the Earth, m the mass of the satellite, r the radius of the satellite's orbit, ω the satellite's angular velocity, and T the orbital period, $T = 24 \times 60 \times 60 = 86\,400$ s. $\frac{GMm}{r^2} = m(r\omega^2) = mr\left(\frac{2\pi}{T}\right)^2 \text{ (show equation)}$ $r^3 = \frac{GMT^2}{4\pi^2}$ The radius of the orbit of the satellite about the centre of the Earth is $r = \sqrt[3]{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(24 \times 60 \times 60)^2}{4\pi^2}} \text{ (values entered explicitly)}$ $= 4.230 \times 10^7 \text{ m}$ Let h be the altitude of the satellite above the surface of the Earth $h = 4.23 \times 10^7 - 6.4 \times 10^6 \text{ (show values)}$ $= 3.59 \times 10^7 \text{ m}$	B1 B1 B1 A0
			2. Linear speed = $r\omega$ $v = r\omega = r\left(\frac{2\pi}{T}\right)$ $= (4.230 \times 10^7)\left(\frac{2\pi}{24 \times 60 \times 60}\right)$ $= 3080 \text{ m s}^{-1}$	B1 A1

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